

Fluid Flow Operation

Text Books:

- **K.A. Gavanhe, “Unit Operations - I”, Nirali Prakashan, 2015**
- W. L. McCabe, J.C. Smith and P. Harriott, “Unit operations of Chemical Engineering”, McGraw Hill, International Edn., 7th Edition, 2005.

Unit 2: Momentum Balance and their Applications

- Kinematics of fluid flow
- Newtonian and non-Newtonian fluids
- Reynolds number - experiment and significance
- Momentum balance
- Forces acting on stream tubes
- Bernoulli's equation
- Correction for fluid friction

Kinematics of fluid flow

- To investigate fluids in motion.
- Fluid mechanics is a science concerned with the response of fluids to forces exerted upon them.
- Hydrostatics: science of water at rest, and hydrodynamics: science of water in motion.
- Archimedes founded hydrostatics in about 250 B.C.
- The foundations of hydrodynamics were laid in the 18th century when mathematicians such as Leonhard Euler and Daniel Bernoulli began to explore a virtually continuous medium like water.
- Their work was continued in the 19th century by several mathematicians and physicists, notably G.G. Stokes and William Thomson.
- By the end of the century explanations had been found for a host of intriguing phenomena having to do with the flow of water through tubes and orifices, the waves that ships moving through water leave behind them, raindrops on windowpanes, etc.

Kinematics of fluid flow

- In fluid mechanics, the state of an isotropic fluid may be completely described by defining its mean mass per unit volume, or density (ρ), its temperature (T), and its velocity (v) at every point in space.
- There are two sorts of stress that may exist in any solid or fluid medium: pressure and shear stress.
- Fluids, by definition, yield to shear stresses no matter how small these stresses may be. They do so at a rate determined by the fluid's viscosity.
- The property viscosity is a measure of the friction that arises when adjacent layers of fluid slip over one another.
- It follows that the shear stresses are everywhere zero in a fluid at rest and in equilibrium, and from this it follows that the pressure (that is, force per unit area) acting perpendicular to all planes in the fluid is the same irrespective of their orientation (Pascal's law).

Kinematics of fluid flow

- For an isotropic fluid in equilibrium there is only one value of the local pressure (p) consistent with the stated values for ρ and T .
- These three quantities are linked together by what is called the equation of state for the fluid.
- For gases at low pressures the equation of state is simple, where R is the universal gas constant (8.3 joules per degree Celsius per mole) and M is the molar mass, or an average molar mass.

$$p = \left(\frac{RT}{M} \right) \rho$$

- For other fluids for which equation of state is not known, the density changes when the pressure is changed by a small amount or the compressibility of the fluid should be known.

Types of Fluid Flow

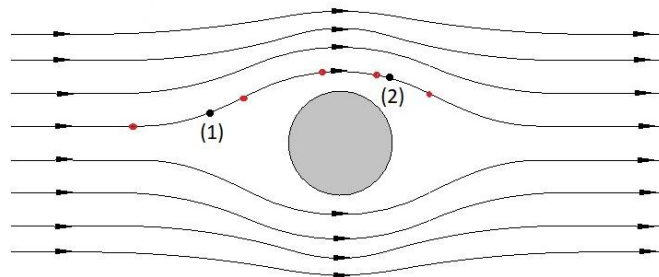
Steady and Unsteady Flow:

- The flow is said to be **steady** if it **does not vary with time**, i.e., the mass flow rate is constant and the quantities, such as temperature, pressure, etc. are independent of time, i.e., do not vary with time.
- If the mass flow rate and/or other quantities such as temperature, pressure, etc. **vary with time, the flow is said to be unsteady.**

Types of Fluid Flow

Stream Line:

- The path of imaginary particles suspended in the fluid and carried along with it.
- In steady flow, the fluid is in motion but the streamlines are fixed.
- A **streamline** is a line that is tangential to the instantaneous velocity direction.
- Where streamlines crowd together, the fluid speed is relatively high; where they open out, the fluid is relatively still.



<https://www.britannica.com/science/streamline>

https://www.princeton.edu/~asmits/Bicycle_web/streamline.html

<https://sbainvent.com/fluid-mechanics/streamlines-and-newtons-second-law/>

Types of Fluid Flow

Stream Tube:

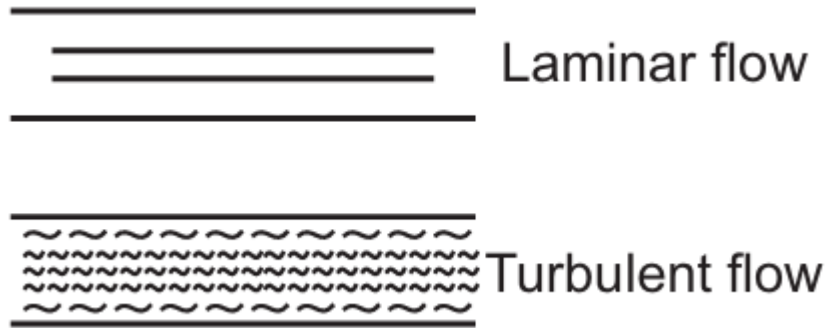
- A stream tube is a tube of small or large cross-section which is entirely bounded by stream lines.
- It may be of any convenient cross-sectional shape and no net flow occurs through the walls of the stream tube.

Types of Fluid Flow

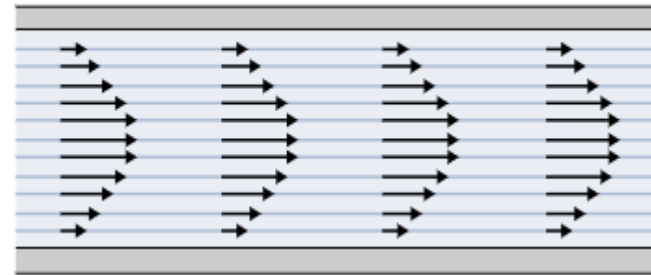
Laminar and Turbulent Flow:

- When a fluid flows steadily through a conduit, either of two different types of flow may occur according to the conditions of flow.
- **The flow in which the streamlines remain distinct/separated from one another over their entire length of flow is termed as laminar flow.**
- This flow is also called as **stream line flow or viscous flow.**
- It is characterised by the **absence of lateral mixing, cross currents and eddies.**
- The flow in which the fluid flows in **parallel, straight lines** is called laminar flow. This occurs at **low fluid velocities.**
- The flow in which the fluid instead of flowing in an orderly manner, **moves erratically in the form of cross currents and eddies is called turbulent flow.**
- This occurs at **high fluid velocities and there is a lateral mixing** in this type of flow.

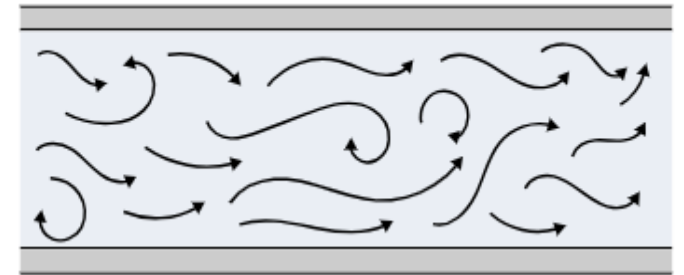
Types of Fluid Flow



Laminar flow



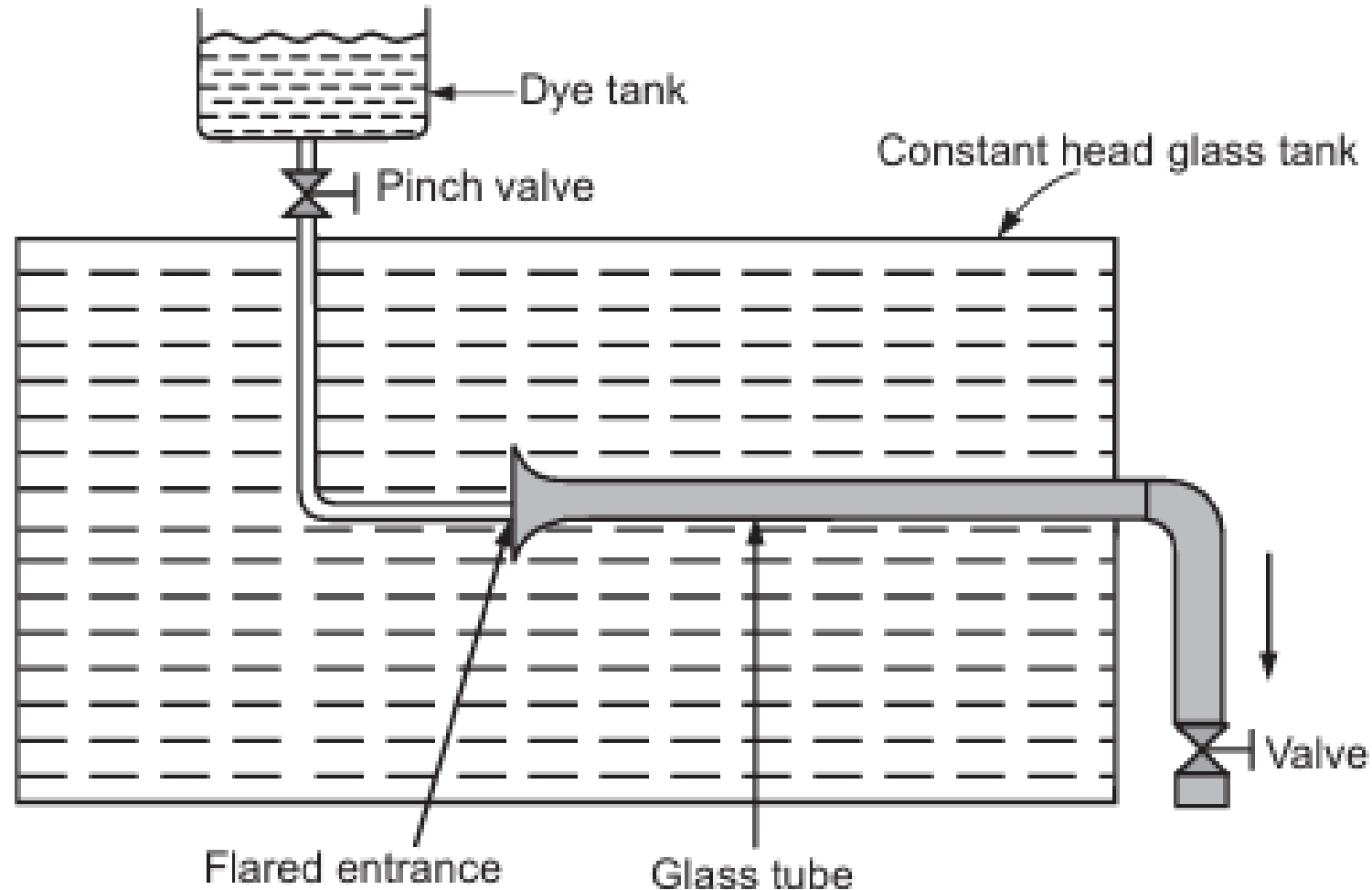
Turbulent flow



Reynolds number - experiment and significance

- This experiment is still used to demonstrate the distinction between the two types of flow, namely laminar flow and turbulent flow.
- It was first conducted by Sir Osborne Reynolds in 1883.
- The experimental set consists of a horizontal glass tube with a flared entrance immersed in a glass walled constant head tank filled with water.
- The flow of water through the glass tube can be adjusted to any desired value by means of a valve provided at the outlet of the tube.
- A capillary tube connected to a small reservoir containing water soluble dye is provided at the centre of the flared entrance of the glass tube for injecting a dye solution in the form of a fine or thin filament into the stream of water.

Reynolds number - experiment and significance



Reynolds number - experiment and significance

- By introducing a water soluble dye into the flow of water, the nature of flow could be observed.
- At low flow rates (i.e., at low water velocities), the filament/thread of coloured water flowed along with the stream of water in a thin, straight line without any lateral mixing.
- This indicated that the water was flowing in the form of parallel streams which did not interfere with each other (i.e., the water was flowing in parallel, straight lines).
- This type of flow pattern is called streamline or laminar.
- The laminar flow is characterised by the absence of bulk movement at right angles to the main stream direction (lateral movement).

Reynolds number - experiment and significance

- As the water flow rate was increased, a definite velocity called the critical velocity was reached, oscillations appeared in the coloured filament/thread and the thread of coloured water became wavy, gradually disappeared and the entire mass of water in the tube became uniformly coloured.
- In other words, the individual particles instead of flowing in an orderly manner parallel to the axis of the tube, moved erratically in the form of cross-currents and eddies which resulted into complete mixing.
- This type of flow pattern is known as turbulent.
- The turbulent flow is characterised by the rapid movement of fluid in the form of eddies in random directions across the tube.
- In between these two types of flow, there exists a transition region wherein the oscillations in the flow were unstable and any disturbance would quickly disappear.

Reynolds number - experiment and significance

$$N_{Re} = \frac{Du\rho}{\mu} \quad \dots (7.1)$$

where

D = diameter of the pipe, m

u = average velocity, m/s

ρ = density of the fluid, kg/m³

and

μ = viscosity of the fluid, kg/(m.s) = (N.s)/m² = Pa.s

For flow in a pipe, the inertia force is proportional to ρu^2 and the viscous force is proportional to $\mu.u/D$.

Reynolds number - experiment and significance

$$\begin{aligned}\frac{\text{Inertia force}}{\text{Viscous force}} &= \frac{\rho u^2}{\mu u/D} \\ &= \frac{Du\rho}{\mu} = N_{Re}\end{aligned}$$

Thus, Reynolds number is the *ratio of the inertia force to the viscous force*. This is an important physical significance of the Reynolds number.

Reynolds number is a useful tool to determine the nature of flow, whether laminar or turbulent.

Reynolds number - experiment and significance

- The Reynolds number is a basic tool to predict the flow pattern in a conduit and is of a vital importance in the study of fluid flow.
- When the value of the Reynolds number is less than 2100, the flow is always laminar and when the value of the Reynolds numbers is above 4000, the flow is always turbulent.
- For Reynolds numbers between 2100 and 4000, a transition region exists and in this region the flow is changing rapidly from laminar to turbulent.

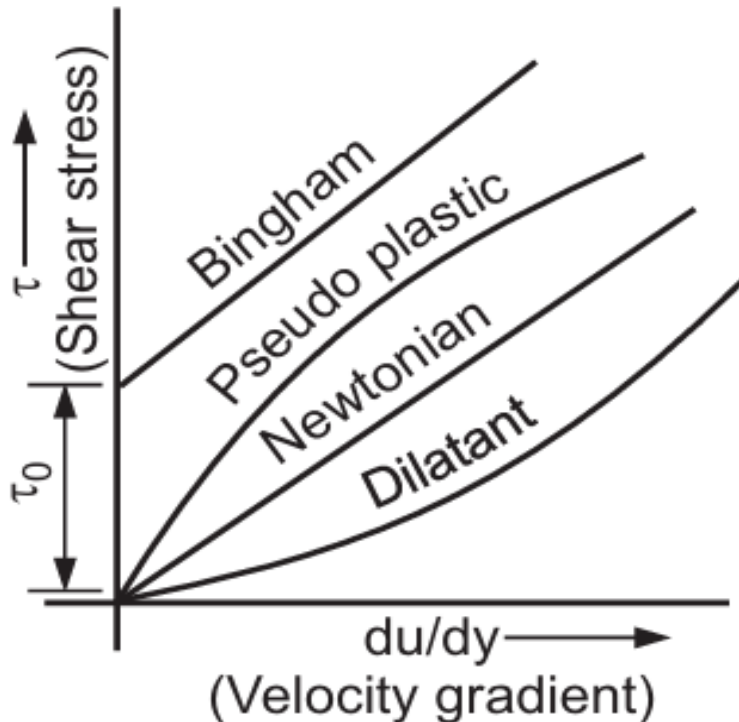
Types of Fluid Flow

- **Potential Flow:** The flow of incompressible fluids without the presence of shear is referred to as potential flow.
- In potential flow, eddies and cross currents cannot form within the stream and friction cannot develop.
- **Fully Developed Flow:** The flow with unchanging velocity distribution is called fully developed flow.

Types of fluids

Newton's Law of Viscosity

$$\tau = \mu \cdot \frac{du}{dy}$$

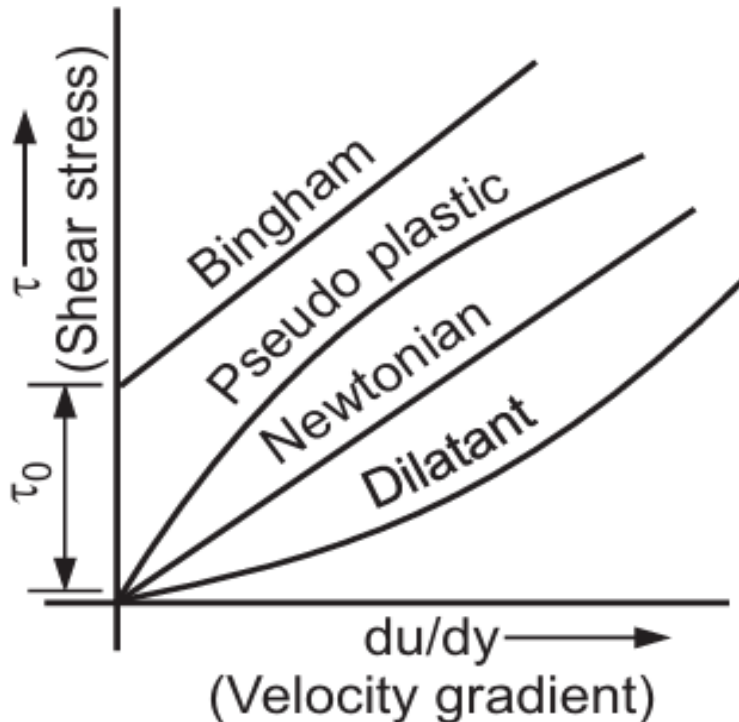


- **Newtonian fluids:**
- **Fluids that obey Newton's law of viscosity, ratio of the shear stress to the rate of shear or shear rate is constant, are called as Newtonian fluids.**
- This is true for all gases and for most pure liquids.
- Plot of τ vs. du/dy results in a straight line passing through the origin.
- Examples: kerosene, alcohol, glycerine, benzene, hexane ether etc., solutions of inorganic salts and of sugar in water.

Types of fluids

Newton's Law of Viscosity

$$\tau = \mu \cdot \frac{du}{dy}$$

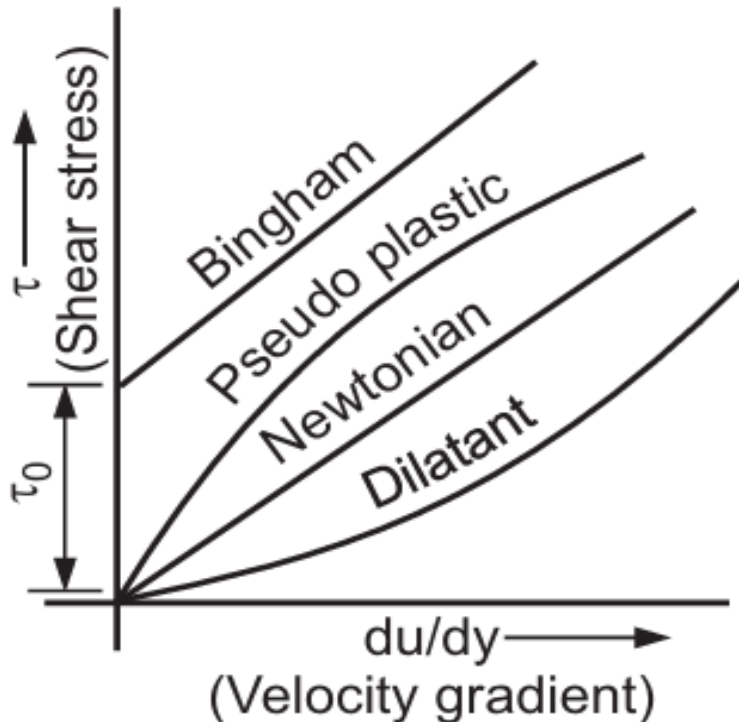


- **Non-Newtonian fluids:**
- Fluids which do not follow Newton's law of viscosity are called as non-Newtonian fluids.
- **Fluids for which the ratio of the shear stress to the shear rate is not constant but is considered as a function of rate of shear.**
- Liquids particularly those containing a second phase in suspension (solutions of finely divided solids and liquid solutions of large molecular weight materials).
- Examples: Tooth pastes, paints, gels, jellies, slurries and polymer solutions.

Types of fluids

Newton's Law of Viscosity

$$\tau = \mu \cdot \frac{du}{dy}$$



- **Non-Newtonian fluids:**

- (a) **Bingham Fluids / Plastics :**

- These fluids **do not deform (flow) unless a threshold shear stress value (yield stress, τ_0) is not exceeded**, then flows linearly under the action of larger shear stress.
 - η is commonly called as the coefficient of rigidity.

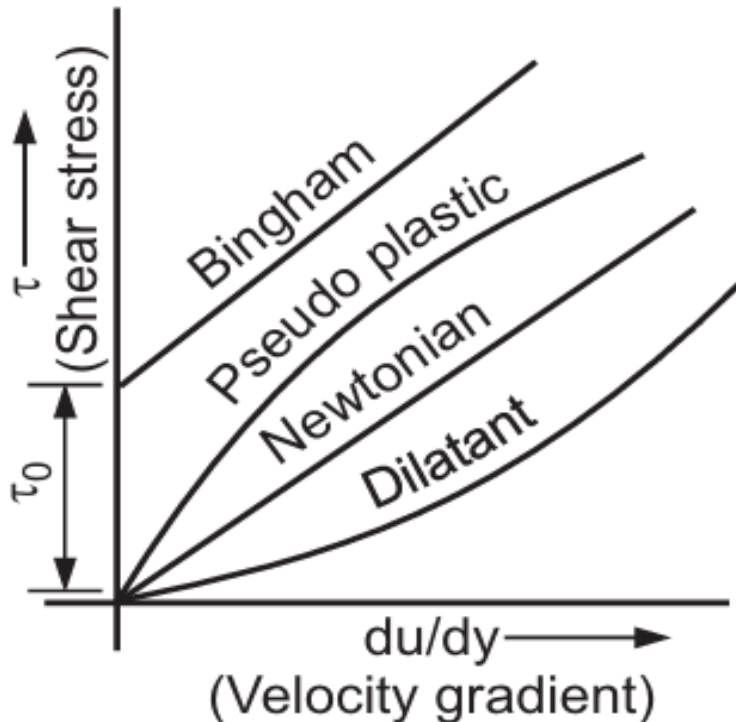
$$\tau = \tau_0, \quad du/dy = 0, \quad \tau > \tau_0, \quad \tau = \tau_0 + \eta \cdot du/dy$$

- Examples : Tooth paste, jellies, paints, sewage sludge and some slurries.

Types of fluids

Newton's Law of Viscosity

$$\tau = \mu \cdot \frac{du}{dy}$$



- **Non-Newtonian fluids:**

- (b) **Pseudoplastic Fluids : Shear thinning fluids**

- **The viscosity of these fluids decreases with increase in velocity gradient/ shear rate.**
 - Examples : Blood, solution of high molecular weight polymers, paper pulp, muds, most slurries and rubber latex.

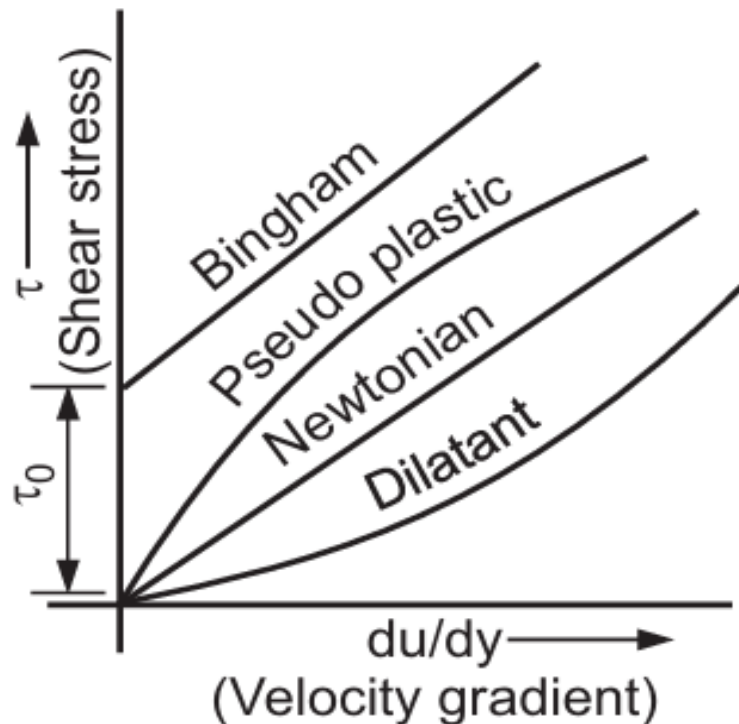
- (c) **Dilatant Fluids : Shear thickening fluids**

- **The viscosity of these fluids increases with an increase in velocity gradient.**
 - Examples : Suspensions of starch in water, pulp in water, and sand filled emulsions.

Types of fluids

Newton's Law of Viscosity

$$\tau = \mu \cdot \frac{du}{dy}$$



- **Non-Newtonian fluids:**
- The experimental curves for pseudoplastic and dilatant fluids can be represented by a **power law called the Ostwald-de-Waele equation.**
- where k and n are arbitrary constants.

$$\tau = k \left(\frac{du}{dy} \right)^n$$

Newtonian fluids : $n = 1$, $k = \mu$

Pseudoplastic fluids : $n < 1$

Dilatant fluids : $n > 1$

Mass Flow rate

- When A is expressed in m^2 , density in kg/m^3 and velocity in m/s , the unit of mass flow rate is kg/s .
- $\dot{m} = \rho u A = \rho (\text{kg}/\text{m}^3) u (\text{m}/\text{s}) A (\text{m}^2)$
- $\dot{m} = \rho u A, \text{kg}/\text{s}$

Average Velocity

- When the flow is not potential flow, the local velocity of fluid will vary from point to point within a given single cross-section of the conduit (tube/pipe).
- Hence, for all practical purposes, it is very convenient to express the average velocity.
- The average velocity (u) of the entire fluid stream flowing through the cross-sectional area A is defined by $u = \dot{m} / (\rho \cdot A)$
- If $\dot{m}$ is expressed in kg/s, ρ in kg/m³ and A in m², then u will be having the units of m/s.
- The average velocity is also equal to the ratio of the volumetric flow rate to the cross sectional area of the conduit.
- $u = Q / A$
- where Q is the volumetric flow rate.
- If Q is expressed in m³/s and A in m², then u will have the units of m/s.
- The mass flow rate through the conduit is related to the volumetric flow rate by the relation $\dot{m} = Q \cdot \rho$

Mass Velocity

- In the case of flow of compressible fluids, it is a usual practice to use the mass velocity.
- The mass velocity does not depend upon temperature and pressure when the flow is steady.
- The mass velocity of a fluid is the ratio of the mass flow rate of the fluid to the cross sectional area of the conduit.
- Mass velocity = $G = \dot{m} A$
- If $\dot{m}$ is expressed in kg/s and A in m^2 , then G will have the units of kg/(m.s).

Equation of Continuity

- It is a mathematical expression for the law of conservation of mass.
- According to the law of conservation of mass for a steady flow system, the rate of mass entering the flow system is equal to that leaving as accumulation is either constant or nil in the flow system under steady conditions.

Consider a flow system (a stream tube of varying cross-section) as shown in Fig. 7.9.

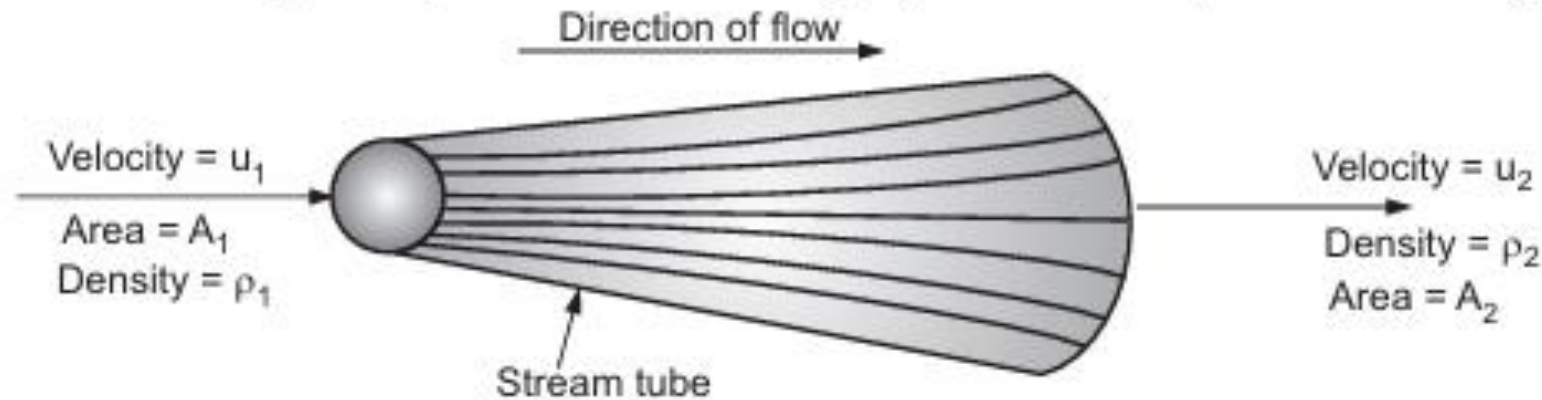


Fig. 7.9 : Continuity

Equation of Continuity

- As the flow cannot take place across the walls of the stream tube, the rate of mass entering the stream tube must be equal to that leaving.
- Let u_1 , ρ_1 and A_1 be the average velocity of the fluid, the density of the fluid and cross-section area of the tube at the entrance, and let u_2 , ρ_2 and A_2 be the corresponding quantities at the exit of the tube.
- Assume that the flow to be potential flow and the density to be constant in a single cross-section.
- Rate of mass entering the flow system = $\rho_1 u_1 A_1$
- Rate of mass leaving the flow system = $\rho_2 u_2 A_2$
- Let $\dot{m}$ be the rate of flow in mass per unit time (mass flow rate of the flowing fluid).

Equation of Continuity

- Under steady flow conditions, according to the law of conservation of mass, the mass of fluid entering the tube in unit time is the same as that leaving the tube. Therefore,

$$\dot{m} = \rho_1 u_1 A_1 = \rho_2 u_2 A_2$$

- It follows for a stream tube, **$\dot{m} = \rho uA = \text{constant}$**
- This equation is known as the equation of continuity. It is applicable to compressible as well as to incompressible fluids.
- In the case of incompressible fluids, $\rho_1 = \rho_2 = \rho$.

Applications of Equation of Continuity

- This equation is used in fluid dynamics to determine the liquid flow in pipes, tubes, channels, and other systems.
- This equation is used in industrial areas to determine the flow of liquids in industrial process.
- This equation is also used to determine the flow rate of water in streams, rivers, groundwater systems.

Equation of Continuity

- The equation of continuity is useful for calculating the velocity of a fluid flowing through pipes of different diameters.
- Assume that we know the velocity (u_1) of a fluid through a pipe of diameter D_1 and we have to obtain the velocity of the fluid (u_2) through a pipe of diameter D_2 which is connected to the pipe of diameter D_1 .
- Then, from the equation of continuity $\dot{m} = \rho_1 u_1 A_1 = \rho_2 u_2 A_2$
- but $\rho_1 = \rho_2$
- where $\dot{m} = u_1 A_1 = u_2 A_2$
- A_1 = cross-section area of the pipe of diameter $D_1 = \pi/4 D_1^2$
- A_2 = cross-section area of the pipe of diameter $D_2 = \pi/4 D_2^2$

$$u_1 (\pi/4.D_1^2) = u_2 (\pi/4.D_2^2)$$
$$u_2 = u_1 (D_1^2 / D_2^2)$$

Momentum balance

Steady flow momentum equation:

- Fluids, either in a static or dynamic motion state, impose forces on immersed bodies and confining boundaries.
- Possible to estimate the steady overall forces applied to either immersed bodies or contained boundaries using basic momentum considerations expressed in Newton's second law.
- Newton's second law states that: The net force acting on a body or system of bodies in a particular direction is equal to the rate of change of momentum of the body or system of bodies in that direction.
- The mathematical formulation of Newton's second law:
- $F = \frac{dP}{dt}$
- Where F is the net force acting on system
- P is total linear momentum of particles in system, $P = m \times v$

https://www.ucl.ac.uk/~uceseug/Fluids2/Notes_Momentum.pdf

Momentum balance

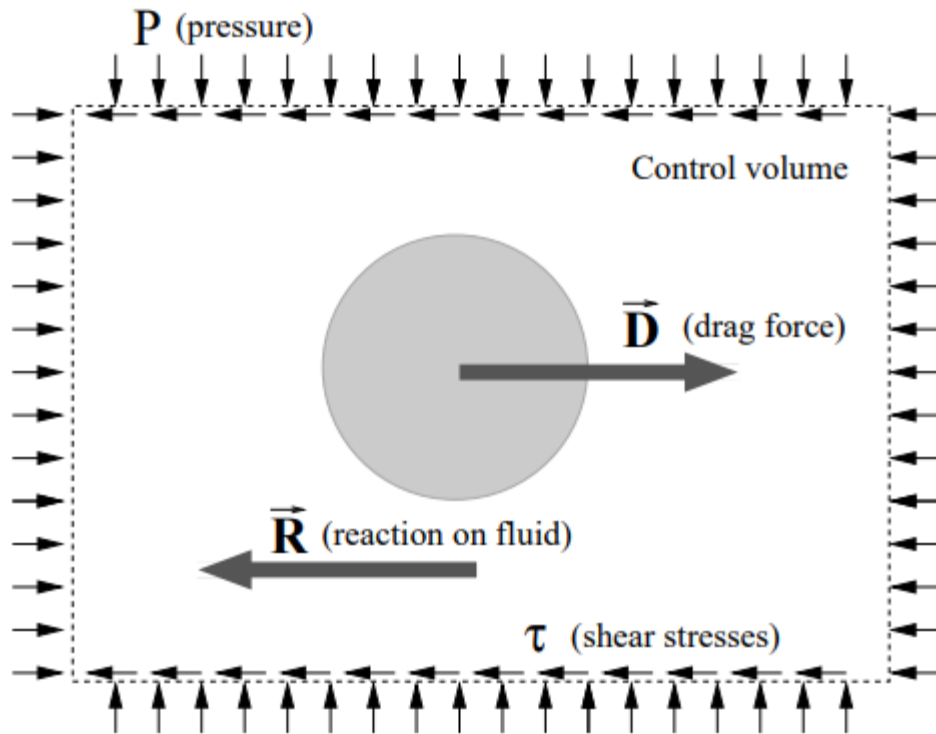
Steady flow momentum equation:

- Newton's second law applied to a control volume.
- **The net force acting on the system is equal to the sum of all external forces acting on the individual particles of the system.**
- Internal forces, that is the forces acting between different particles of the system, do not contribute to the change of the momentum of the system because by Newton's third law for each such a force there is an equal opposite reaction.
- **For a fluid it is convenient to apply physical principles to a system of fluid particles occupying a certain suitable volume called a control volume.**

Momentum balance

Steady flow momentum equation:

- Newton's second law applied to a control volume.



- The force on fluid particles within control volume include:
 1. Gravity or inertial forces
 2. Reaction from the body drag D
 3. Pressure or shear forces over the bounding surface of the control volume plus any weight components

https://www.ucl.ac.uk/~uceseug/Fluids2/Notes_Momentum.pdf

Momentum balance

Steady flow momentum equation:

The force F_T is made up of three components:

✓ $F_1 = F_R$ = Force exerted in the direction of the fluid by any **solid body** touching the control volume.

✓ $F_2 = F_B$ = Force exerted in the direction of the fluid by a **body force** (gravity force).

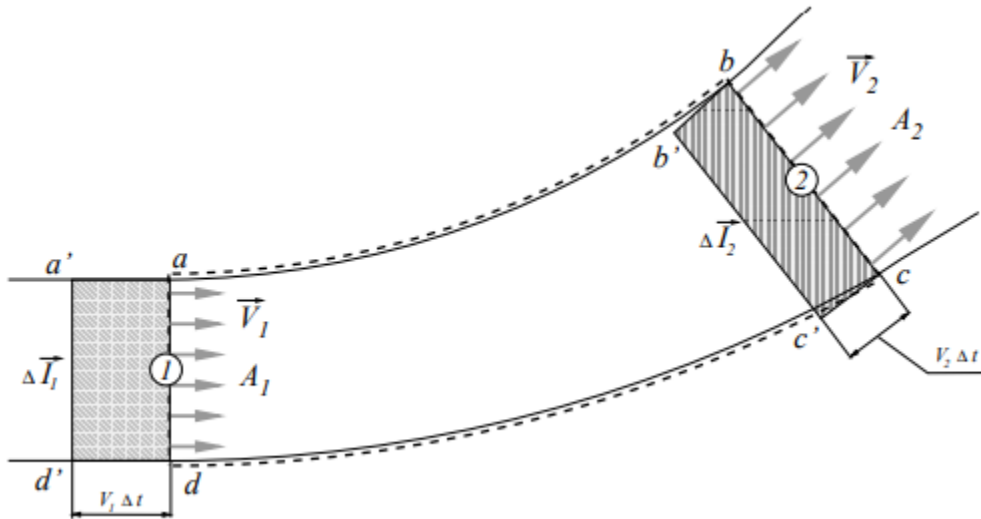
$$F_B = \text{Fluid weight} + \text{object weight} = (\rho V g)_{\text{fluid}} + (\rho V g)_{\text{object}}$$

✓ $F_3 = F_P$ = Force exerted in the direction of the fluid by a **fluid pressure** outside the control volume

$F_P = \vec{F}_{P,1} + \vec{F}_{P,2} + \cdots \vec{F}_{P,n}$ (If the flow is in two dimensional the pressure will be analyze in x and y directions).such that $F_P = \text{Pressure} \times \text{Area}$

Momentum balance

Steady flow momentum equation:

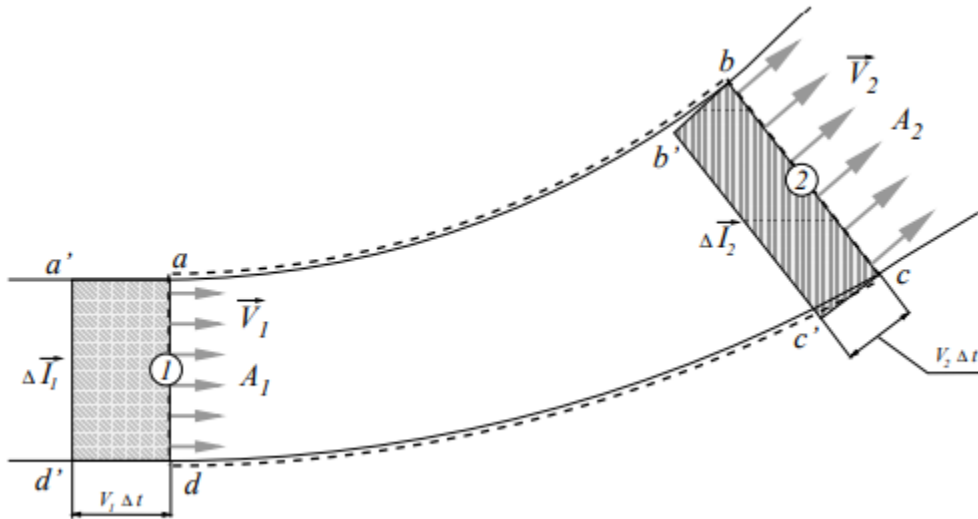


- Let us consider a stream tube of a small cross section such that the fluid velocity can be assumed constant across the tube.
- Part of stream tube between two cross section 1 and 2 taken as a control volume (region abcd).
- The areas of the cross section 1 and 2 are $A_{1,2}$ and the corresponding fluid velocity is $V_{1,2}$.
- During a small time interval Δt the fluid in the region $a'dd'$ enters the control volume, and the corresponding mass gain is

$$\Delta m_1 = \rho A_1 V_1 \Delta t.$$

Momentum balance

Steady flow momentum equation:



- Fluid in region bcc'd' exits the control volume , and mass loss is

$$\Delta m_2 = \rho A_2 V_2 \Delta t.$$

- According to equation of continuity, mass flow rates across surface 1 and 2 are equal so

$$\Delta m_1 = \Delta m_2 = \rho Q \Delta t,$$

- Where Q is volumetric flow rate

Momentum balance

Steady flow momentum equation:

- Momentum for cross section 1 : $\Delta P_{1,2} = \rho Q \Delta t v_1$
- Momentum for cross section 2 : $\Delta P_{1,2} = \rho Q \Delta t v_2$
- Momentum for flow of masses $\Delta m_{1,2}$
 $\Delta P_{1,2} = \rho Q \Delta t v$
- For steady flow momentum of fluid inside control volume does not change $\Delta P = 0$.
- This implies that certain force should act on fluid inside control volume to compensate change of momentum due to fluid flow in and out of control volume.
- According to Newton's second law: change in momentum caused by an action of such a force during time period Δt is: $\Delta P = F \Delta t$

Momentum balance

Steady flow momentum equation:

- Overall balance of change of momentum in control volume is

$$\Delta P = \rho Q \Delta v_1 \Delta t - \rho Q \Delta v_2 \Delta t + F \Delta t = 0$$

- Total force on fluid in control volume given by

$$F = \rho Q \Delta v_2 - \rho Q \Delta v_1 = \rho Q (v_2 - v_1)$$

- **Momentum principle:** For a steady flow the total force on the fluid in the control volume is equal to the momentum flow rate out of the control volume minus the momentum flow rate into the control volume.

Momentum balance

Steady flow momentum equation:

The final relationship for momentum equation will be as following:

$$\mathbf{F}_T = \mathbf{F}_R + \mathbf{F}_B + \mathbf{F}_P = \rho Q(\mathbf{v}_2 - \mathbf{v}_1)$$

$\mathbf{F}_R$ = The force exerted by the **body** on the **fluid**

But, $\mathbf{R} = -\mathbf{F}_R$ = The force exerted by the **fluid** on the **body**

Usually the value of $\mathbf{R}$ is required for design purposes :

$$\mathbf{F}_T = -\mathbf{R} + \mathbf{F}_B + \mathbf{F}_P = \rho Q(\mathbf{v}_2 - \mathbf{v}_1) \rightarrow \mathbf{R} = \mathbf{F}_B + \mathbf{F}_P - \rho Q(\mathbf{v}_2 - \mathbf{v}_1)$$

$$\rightarrow \mathbf{R} = \mathbf{F}_B + \mathbf{F}_P + \rho Q(\mathbf{v}_1 - \mathbf{v}_2) = \text{Resultant force acting on the body}$$

Applications of momentum balance

- Momentum balance, a form of Newton's second law, is applied in fluid mechanics to calculate forces on structures and to analyze flow behavior by relating the forces acting on a fluid to its momentum change.
- Key applications include calculating forces on pipe bends and fittings, analyzing open channel flows such as hydraulic jumps, and designing pumps and turbines.
- The momentum equation helps determine forces and pressure distributions in various fluid systems by considering changes in velocity, area, and pressure during flow.

Bernoulli's equation

- An important relation, called the Bernoulli equation without friction can be derived on the basis of Newton's second law of motion (force is equal to the rate of change of momentum) for potential flow.
- **It is simply an energy balance.**
- The relation that will be obtained is strictly applicable to an **inviscid (frictionless) fluid**.
- Let us consider an element of length ΔL of a stream tube of constant cross-sectional area A and the density of the fluid be ρ .
- Force acting in the direction of flow is taken as positive.

Bernoulli's equation

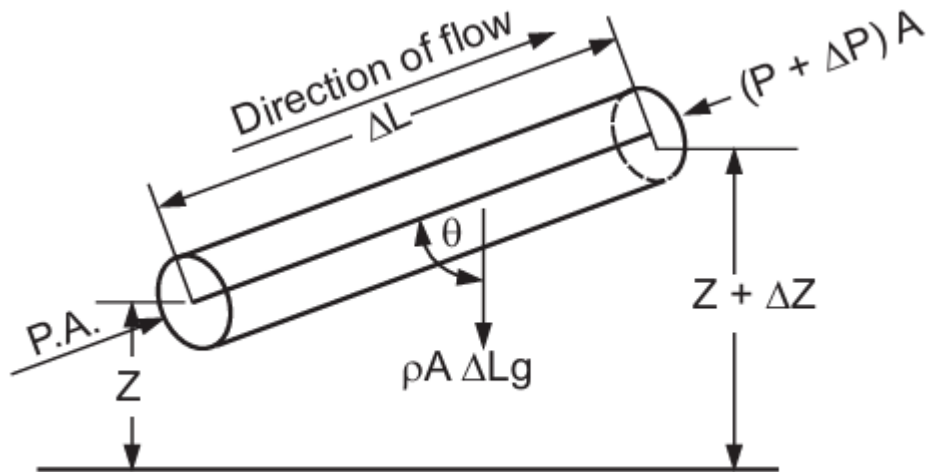


Fig. 7.11 : Force balance for potential flow

The forces acting on the element (treating the element as a free body) are:

1. The force from the upstream pressure = PA (positive)
2. The force from the downstream pressure normal to the cross-section of the tube = $(P + \Delta P) A$ (negative)
3. The force from the weight of fluid (negative) = $\rho A \Delta L g$

The component of this force acting opposite to the direction of flow is $\rho A \Delta L g \cos \theta$.

The 1st force helps the flow while the remaining two forces oppose the flow.

Bernoulli's equation

According to the Newton's second law of motion,

$$\left\{ \begin{array}{l} \text{Sum of all forces} \\ \text{acting in the direction of flow} \end{array} \right\} = \left\{ \begin{array}{l} \text{Rate of change of} \\ \text{momentum of a fluid} \end{array} \right\}$$

$$PA - (P + \Delta P) A - \rho A \Delta L g \cos \theta = \rho \cdot u \cdot A \Delta u$$

$$- \Delta P \cdot A - \rho A \Delta L g \cos \theta = \rho \cdot u \cdot A \Delta u$$

$$\Delta P \cdot A + \rho A \Delta L g \cos \theta + \rho \cdot u \cdot A \Delta u = 0$$

Dividing each term of Equation by $A \cdot \Delta L \cdot \rho$

$$\frac{\Delta P}{\rho \Delta L} + g \cos \theta + \frac{u \cdot \Delta u}{\Delta L} = 0$$

Bernoulli's equation

Keeping $\cos \theta = \Delta Z / \Delta L$

$$\frac{1}{\rho} \frac{\Delta P}{\Delta L} + g \frac{\Delta Z}{\Delta L} + u \frac{\Delta u}{\Delta L} = 0$$

In the differential form

$$\frac{1}{\rho} \frac{dP}{dL} + g \frac{dZ}{dL} + \frac{d(u^2/2)}{dL}$$

$$\frac{dP}{\rho} + g dZ + d(u^2/2) = 0$$

Bernoulli's equation

For incompressible fluids, density is independent of pressure

On integration,

$$\frac{P}{\rho} + gZ + \frac{u^2}{2} = \text{constant}$$

Equation is known as the **Bernoulli equation**.

The Bernoulli's equation relates the pressure at a point in the fluid to its position and velocity.

Each term in the Bernoulli equation represents **energy per unit mass of the fluid** and has the units of J/kg in the SI system.

Bernoulli's equation

The unit of $\frac{P}{\rho}$ is : $\left(\frac{\text{N}}{\text{m}^2}\right) \times \frac{1}{(\text{kg}/\text{m}^3)} = \frac{\text{N.m}}{\text{kg}} = \text{J/kg}$

The unit of $g Z$ is : $\left(\frac{\text{m}}{\text{s}^2}\right) (\text{m}) = \left(\frac{\text{kg}}{\text{kg}}\right) \left(\frac{\text{m}}{\text{s}^2}\right) (\text{m}) = \left(\frac{\text{kg.m}}{\text{s}^2}\right) \left(\frac{\text{m}}{\text{kg}}\right)$
 $= \frac{\text{N.m}}{\text{kg}} = \text{J/kg}$

The unit of $u^2/2$ is : $\frac{\text{m}^2}{\text{s}^2} = \left(\frac{\text{kg}}{\text{kg}}\right) \left(\frac{\text{m}^2}{\text{s}^2}\right) = \left(\frac{\text{kg.m}}{\text{s}^2}\right) \left(\frac{\text{m}}{\text{kg}}\right) = \frac{\text{N.m}}{\text{kg}} = \text{J/kg}$
 $[\text{As } 1 \text{ N} = 1 (\text{kg.m})/\text{s}^2]$

Bernoulli's equation

1. Pressure Energy:

- It is the work which must be done in order to introduce a fluid into a system without change in the volume.
- It is the energy of the fluid due to pressure acting on it.

$$\begin{aligned}\text{Pressure energy (flow energy)} &= \text{work done on a fluid} \\ &= \text{force} \times \text{displacement} \\ &= \frac{\text{force}}{\text{area}} \times \text{area} \times \text{displacement} \\ &= \text{pressure} \times \text{volume}\end{aligned}$$

$$\begin{aligned}\text{Pressure energy per unit mass of fluid} &= \frac{\text{pressure} \times \text{volume}}{\text{mass}} = \frac{\text{pressure}}{\text{mass/volume}} \\ &= \frac{P}{\rho}, \text{ J/kg in the SI system.}\end{aligned}$$

Bernoulli's equation

2. Kinetic Energy:

- It is the energy of a fluid by virtue of its motion with reference to some arbitrarily fixed body.
- The kinetic energy of a fluid of mass m moving with velocity u is given by

$$\text{Kinetic energy} = \frac{1}{2} mu^2$$

$$\text{Kinetic energy per unit mass of fluid} = \frac{1}{2} u^2, \text{ J/kg in the S.I. system}$$

Bernoulli's equation

3. Potential Energy:

It is the energy of a fluid due to its position in the earth's gravitational field. It is equal to the work that must be done on the fluid in order to raise it to a certain position from some arbitrarily chosen datum level.

At a datum level, the potential energy is taken as zero.

The potential energy of a fluid of mass m situated at a height Z above a datum level is given by

$$\text{Potential energy} = mgZ$$

$$\text{Potential energy per unit mass} = gZ, \text{ J/kg in the S.I. system}$$

Bernoulli's equation

Bernoulli's equation can be written in an alternate form by dividing by g

$$\frac{P}{\rho g} + Z + \frac{u^2}{2g} = \text{constant}$$

Each term in this equation represents energy per unit weight of the fluid and has the dimensions of length

$P/\rho g$ is the pressure head / static head, Z is the potential head and $u^2/2g$ is the velocity head/kinetic head.

The sum of the pressure head, velocity head, and potential head is known as the total head or the total energy per unit weight of the fluid.

The Bernoulli equation states that in a steady irrotational flow of an incompressible fluid, the total energy at any point is constant.

Bernoulli's equation

Bernoulli's equation when applied between stations 1 and 2 in the direction of flow becomes

$$\frac{P_1}{\rho} + g Z_1 + \frac{u_1^2}{2} = \frac{P_2}{\rho} + g Z_2 + \frac{u_2^2}{2}$$

The sum of the pressure head and potential head

$P/\rho g + Z$

is termed as the **piezometric head**.

Correction for fluid friction

- The Bernoulli equation is derived for the frictionless fluid.
- Therefore, it must be corrected for the existence of the fluid friction whenever boundary layer forms.
- Fluid friction is an irreversible conversion of mechanical energy into heat.
- Thus, the quantity $P/\rho + u^2/2 + gZ$ is not constant but always decreases in the direction of flow.
- The Bernoulli equation for incompressible fluids is corrected for friction by adding a friction term on the R.H.S. of Equation.
- **h_f is the total frictional loss of energy due to friction** between stations 1 and 2 in J/kg.

$$\frac{P_1}{\rho} + g Z_1 + \frac{u_1^2}{2} = \frac{P_2}{\rho} + g Z_2 + \frac{u_2^2}{2} + h_f$$

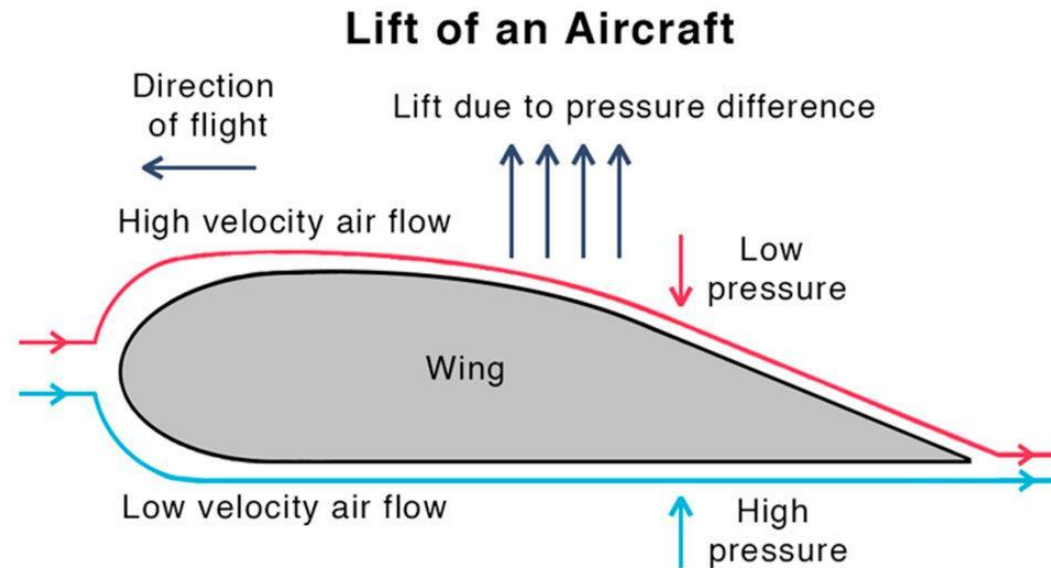
Applications of Bernoulli's equation

- **Aerodynamics:** It explains how airplanes generate lift. The shape of an airplane's wing is designed so that air moves faster over the top surface, creating lower pressure compared to the underside and thus lifting the plane.
- **Venturi Effect:** This principle is used in the design of the Venturi tube, where a fluid's velocity increases as it passes through a constricted section of the tube, leading to a pressure drop. This effect is utilized in carburetors and aspirators.
- **Hydraulic Machinery:** Bernoulli's Theorem underpins the operation of hydraulic pumps and turbines, which convert mechanical energy into fluid energy and vice versa, based on pressure and velocity changes.
- **Sports:** Understanding the theorem helps in analyzing the behavior of balls in sports, explaining phenomena like the curveball in baseball or soccer.
- **Civil Engineering:** In the design of bridges and buildings, Bernoulli's principle helps in understanding the effect of wind pressure on structures, ensuring their stability and safety.

<https://www.geeksforgeeks.org/physics/application-of-bernoullis-theorem/>

Applications of Bernoulli's equation

- **How do airplanes fly?**
- The **Bernoulli Equation** states that a speed increase on moving air or a flowing fluid is accompanied by a decrease in the pressure of said air or fluid. That is, at higher speed lower pressure (and viceversa) to maintain the same value.



Watch Video:

<https://www.youtube.com/watch?v=U-rwTdkuFKM>

<https://worldaviationato.com/en/bernoulli-principle/>